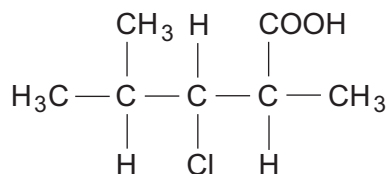


SECTION A

Answer **all** questions in the spaces provided.

1. (a) Name the compound whose formula is shown below. [1]



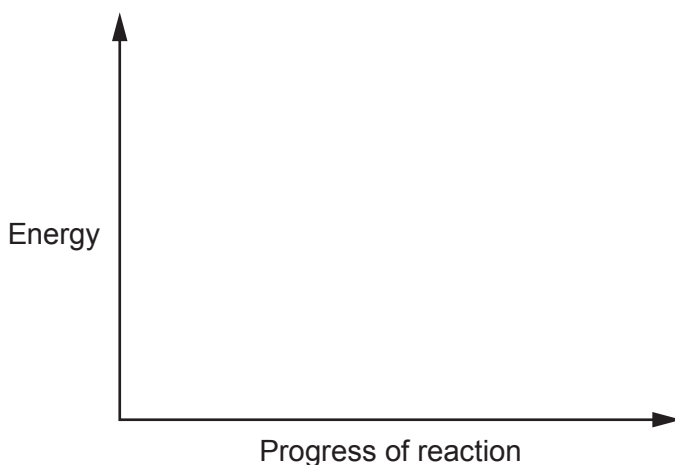
- (b) Draw the skeletal formula of the compound shown in (a). [1]

2. Respiration involves the release of energy by using foods. The equation for the respiration of glucose is shown below.



Photosynthesis is the process by which plants store energy in foods. It can be thought of as the reverse of respiration.

On the axes below sketch and label the energy profile for the **photosynthesis** process. You should label the enthalpy change, ΔH , and the activation energy, E_a . [2]



SECTION A

Answer all questions in the spaces provided.

1. (a) Draw the **displayed** formula of (*E*)-pent-2-ene. [1]

- (b) Draw the **skeletal** formula of 3-chlorohex-1-ene. [1]

2. State the molecular formula of a straight-chained saturated halogenoalkane whose molecules each contain 50 carbon atoms and 2 chlorine atoms. [1]

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SECTION A

Answer **all** questions in the spaces provided.

1. Draw the structure of a secondary alcohol that contains 6 carbon atoms. [1]

2. Draw the **skeletal** formula of 3-chloro-2,2-dimethylpentane. [1]

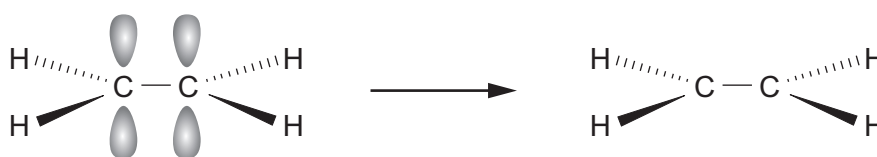
3. Explain why ethanol is soluble in water but ethane is not. [2]

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4. Complete the diagram to show the formation of the π -bond in ethene. [1]



SECTION A

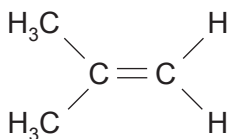
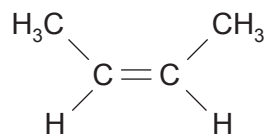
Answer **all** questions in the spaces provided.

1. Draw the **skeletal** formula of 2-bromo-3-methylbutane. [1]

2. Write the equation that corresponds to the standard molar enthalpy change of formation of magnesium carbonate, MgCO_3 . [2]

.....

3. The compounds shown below have the same molecular formula.

**A****B**

State which of these compounds can exist as *E-Z* isomers.

Explain your answer and name the isomer.

[2]

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.....
.....

Name of isomer



SECTION AAnswer **all** questions.

1. Bromine water can be used to test for alkenes.

(a) (i) State the expected colour change for a positive test for alkenes. [1]

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(ii) Draw the structure of the product formed when propene reacts with bromine water. [1]

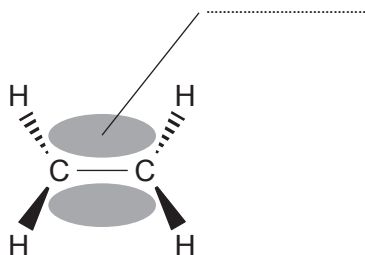
(b) Identify another reagent that can be used to test for the presence of alkenes. [1]

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2. Bonds in hydrocarbons are formed by the overlap of orbitals between each atom.

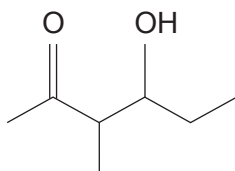
(a) Draw an *s*-orbital and a *p*-orbital in the space below. [1]

(b) Name the type of bond shown in the diagram below. [1]



SECTION AAnswer **all** questions.

1. Give the molecular formula of the compound shown. [1]



Molecular formula

2. When fuels are burned with insufficient oxygen, incomplete combustion occurs and carbon monoxide is formed instead of carbon dioxide.

Write the equation for the incomplete combustion of propane, C_3H_8 . Assume that carbon monoxide is the only carbon-containing product. [1]

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3. Name the reagent that is needed to change unsaturated oils into saturated fats. [1]

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